آغا خان یونیورسٹی ایگزامینیشن بورڈ

AGA KHAN UNIVERSITY EXAMINATION BOARD

Notes from E-Marking Centre on HSSC-II Physics Annual Examinations 2025

Introduction

This document has been produced for the teachers and candidates of the Higher Secondary School Certificate (HSSC) Part II Physics. It contains comments on candidates' responses to the 2025 HSSC-II Examination, indicating the quality of the responses and highlighting their relative strengths and weaknesses.

E-Marking Notes

This includes overall comments on candidates' performance on every question and *some* specific examples of candidates' responses that support the mentioned comments. Please note that the descriptive comments represent an overall perception of the better and weaker responses as gathered from the e-marking session. However, the candidates' responses shared in this document represent some specific example(s) of the mentioned comments.

Teachers and candidates should be aware that examiners may ask questions that address the Student Learning Outcomes (SLOs) in a manner that requires candidates to respond by integrating knowledge, understanding and application skills they have developed during the course of study. Candidates are advised to read and comprehend each question carefully before writing the response to fulfill the demand of the question.

Candidates need to be aware that the marks allocated to the questions are related to the answer space provided on the examination paper as a guide to the length of the required response. A longer response will not in itself lead to higher marks. Candidates need to be familiar with the command words in the SLOs, which contain terms commonly used in examination questions. However, candidates should also be aware that not all questions will start with or contain one of the command words. Words such as 'how', 'why' or 'what' may also be used. It is imperative to refer to the command word guide available on the AKU-EB website for understanding the expectations of the command word.

General Observations

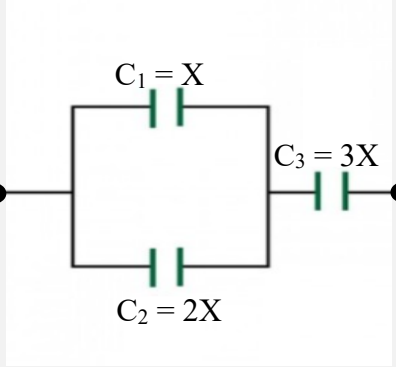
The overall performance of the students was commendable, demonstrating a strong understanding of concepts, particularly in topics involving direct conceptual knowledge and standard applications. Most students exhibited a good command of basic principles in areas such as amplitude and frequency modulations; they have an average command of elastic and plastic deformations, and unsatisfactory performance on forward and reverse bias. In these areas, responses were generally accurate, well-structured, and reflected a sound grasp of both theoretical and practical aspects.

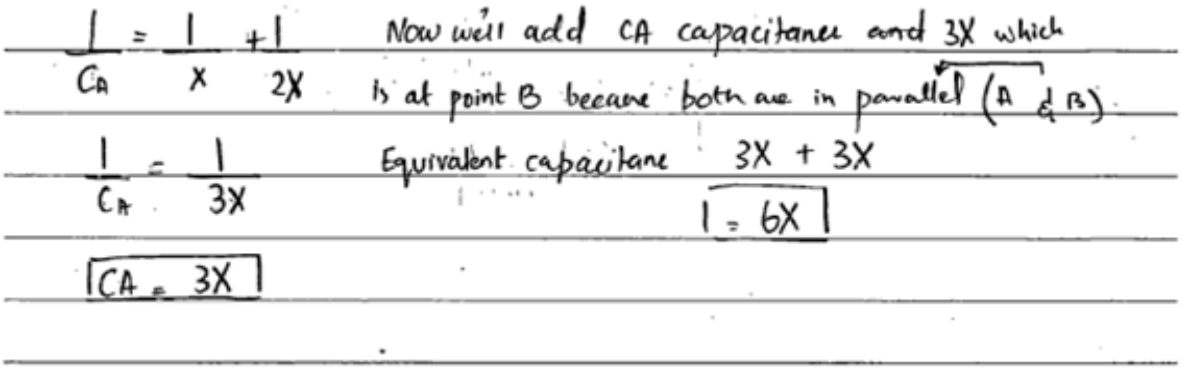
However, certain areas need improvement to ensure a more thorough understanding, such as capacitance, electromagnetism, simple AC and DC generators, DC motors, inner shell transition and characteristics, nuclear reactors, carbon resistance, power dissipation, photoelectric effect, and Compton's effect.

Note: Candidates' responses shown in this report have not been corrected for grammar, spelling, format, or information.

DETAILED COMMENTS

Constructed Response Questions (CRQs)

Question No. 1	
Question Text	Calculate the equivalent capacitance between points A and B. 
SLO No.	11.7.2
SLO Text	Calculate the capacitance of different capacitors in series and in parallel using formulae.
Max Marks	03
Cognitive Level	A*
Checking Hints	1 mark for writing each formula of a series or parallel combination of capacitors 1 mark for finding the correct value of capacitance in series 1 mark for finding the correct value of capacitance in parallel
Overall Performance	The candidates' overall performance on this question was not good. Most were unable to correctly identify the series and parallel parts of the capacitor network and apply appropriate formulas to find the equivalent capacitance.
Description of Better Responses	*Better responses* demonstrated candidates' strong proficiency in analysing capacitor circuits, accurately identifying the parallel combination of C_1 and C_2 (C_{12} = 3X) and its subsequent series connection with C_3. These solutions exhibited precise application of formulas, including the correct series capacitance calculation ($\frac{1}{C_E} = \frac{1}{3X} + \frac{1}{3X}$) and proper simplification to arrive at C_E = 1.5X. The work showed methodical, step-by-step reasoning with clear intermediate steps, reflecting both conceptual understanding and computational accuracy. These responses indicated a strong grasp of both the theoretical concepts and practical problem-solving techniques required for complex circuit analysis.
Image of Better Response	C_1 & C_2 are connected in parallel So $C_{1-2} = C_1 + C_2 \Rightarrow X + 2X = 3X$ $C_{1-2} = 3X$ Now C_{1-2} & C_3 are connected in series: $\frac{1}{C_E} = \frac{1}{C_{1-2}} + \frac{1}{C_3}$ $\frac{1}{C_E} = \frac{1}{3X} + \frac{1}{3X}$ $\frac{1}{C_E} = \frac{1}{3X} + \frac{1}{3X} \Rightarrow \frac{2}{3X}$ $\frac{1}{C_E} = \frac{2}{3X} \Rightarrow C_E = 1/2/3X$ $C_E = \frac{3X}{2} = 1.5X$

Description of Weaker Responses	*Weaker responses* demonstrated difficulties in analysing capacitor circuits, particularly in distinguishing between series and parallel combinations contained fundamental errors, including incorrect application of capacitance formulas, such as treating reciprocal series calculations as parallel additions and mathematical mistakes in simplifying expressions. These mistakes reflect a lack of clarity in applying circuit principles and performing mathematical operations, highlighting the need for reinforced instruction on identifying circuit configurations and executing proper calculations. Such responses indicate significant gaps in understanding that require targeted practice with basic circuit analysis problems. Some candidates solved only part of the question correctly, typically identifying one type of combination but not the other, resulting in incomplete answers. A few responses reflected fundamental misconceptions about circuit arrangement, leading to incorrect combinations and calculation errors.
Image of Weaker Response	

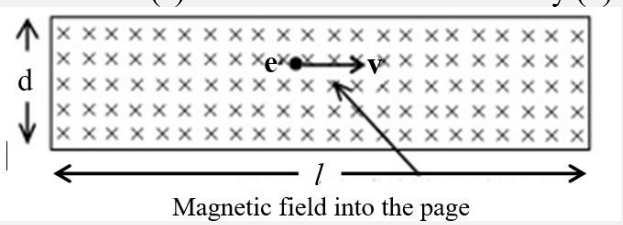
Suggestions for improvement (Highlight all that apply)

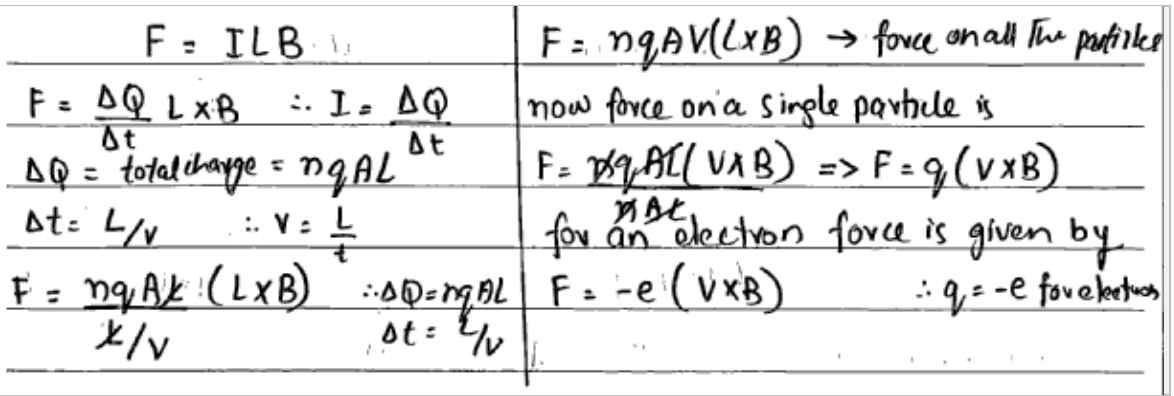
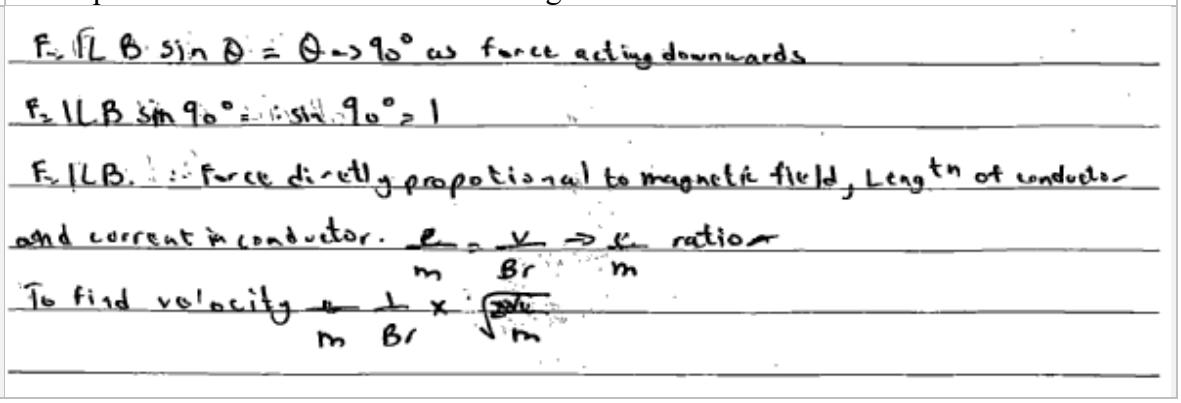
Maximising SLO Achievement	Preferred Pedagogy** Used for this SLO	Assessment Strategies
• Identify the expectation of command words (use Command Word Guide) • Ensure the content is taught at the relevant cognitive level • Identify necessary content required (skills + concepts) • Review past paper questions on the concept • Utilise the resource guide for additional materials	• Story Board • Cause and Effect • Fish and Bone • Concept Mapping • Audio Visual Resources • Think, Pair, and Share • Knowledge Platform videos • Questioning Technique (Socratic approach) • Practical Demonstration ** For description of each Pedagogy, refer to Annexure A	• Past paper questions • Discussion on E-Marking Notes • AKU-EB Digital Learning Solution powered by Knowledge Platform https://akueb.knowledgeplatform.com/login

Any Additional Suggestions:

- Teachers explicitly teach the visual and mathematical distinctions between series and parallel capacitor combinations. Use labelled circuit diagrams and step-by-step breakdowns to reinforce correct identification, as misclassification was a recurring error.
- Emphasise the proper use of reciprocal calculations for series capacitance $\left(\frac{1}{C_{eq}} = \frac{1}{C_1} + \frac{1}{C_2}\right)$ and direct addition for parallel $(C_{eq} = C_1 + C_2)$. Provide practice problems focusing solely on formula transitions to prevent mix-ups.
- Encourage students to follow a structured approach: (1) identify circuit segments, (2) apply correct formulas stepwise, and (3) verify units and physical plausibility of results. This reduces omissions and logical gaps.

*K = Knowledge U = Understanding A = Application and other higher-order cognitive skills

Question No. 2	
Question Text	The given diagram shows a magnetic field normal to a metallic strip of width (d) in which an electron (e) moves with constant velocity (v).  Magnetic field into the page Deduce an equation for the magnetic force applied for the given case
SLO No.	13.2.1
SLO Text	Derive an equation for the force on a moving charge in a uniform magnetic field and a beam of particles and use this equation for solving word problems.
Max Marks	03
Cognitive Level	U
Checking Hints	1 mark for writing the correct formula of F = I L B sinθ 1 mark for writing the correct formula of I = q / t 1 mark for writing F = q v B
Overall Performance	Overall, the cohort were unable to demonstrate a solid grasp of the underlying principles, with most candidates failing to derive the relationship for the magnetic force on a moving charge $(F = q\,v\,B\,\sin\theta)$ by logically connecting it to the magnetic force on a current-arraying conductor $(F = I\,L\,B\,\sin\theta)$, substituting $I = \frac{q}{t}$ and $L = v\,t$ to arrive at the final expression.
Description of Better Responses	*Better responses* effectively demonstrated their understanding by clearly deriving the magnetic force on a moving charge, $F = q\,v\,B\,\sin\theta$ from the current-carrying conductor formula $F = I\,L\,B\,\sin\theta$. They carefully substituted current ($I = \Delta Q / \Delta t$) and total charge ($\Delta Q = n\,q\,A\,L$), then logically simplified using time ($\Delta t = L / v$) to connect macroscopic current with microscopic charge motion. These solutions stood out through their accurate mathematical treatment, including proper use of the cross product ($v \times B$) and clear explanation of key assumptions like charge density ($n\,q\,A\,L$). The good answers went further by applying the derivation to specific cases (e.g., $F = -e\,(v \times B)$ for electrons), showing both solid conceptual grasp and thorough attention to detail.

Images of Better Responses	
Description of Weaker Responses	*Weaker responses* struggled to derive the magnetic force on a moving charge, often confusing it with the force on a current-carrying conductor (F = I L B) without properly transitioning to charge motion (F = q v B). Many introduced irrelevant concepts like e / m ratio or mishandled the role of the angle in the cross-product, while others simply stated the final formula without showing the necessary derivation steps. Additionally, the provided diagram of electron motion was frequently ignored, leading to missed opportunities to justify direction (v × B) or charge polarity (q = - e). These errors revealed deeper issues: weak conceptual links between macroscopic and microscopic physics, poor problem structure (e.g., missing derivation steps), and underutilisation of visual information. A smaller group accurately derived the force for a current-carrying conductor but did not extend the logic to isolate the charge, indicating a partial understanding of the conceptual link between current and charge motion.
Images of Weaker Responses	

Suggestions for improvement (Highlight all that apply)

Maximising SLO Achievement	Preferred Pedagogy Used for this SLO	Assessment Strategies
• Identify the expectation of command words (use Command Word Guide) • Ensure the content is taught at the relevant cognitive level • Identify necessary content required (skills + concepts) • Review past paper questions on the concept • Utilise the resource guide for additional materials	• Story Board • Cause and Effect • Fish and Bone • Concept Mapping • Audio Visual Resources • Think, Pair and Share • Knowledge Platform videos • Questioning Technique (Socratic approach) • Practical Demonstration	• Past paper questions • Discussion on E-Marking Notes • AKU-EB Digital Learning Solution powered by Knowledge Platform https://akueb.knowledgeplatform.com/login

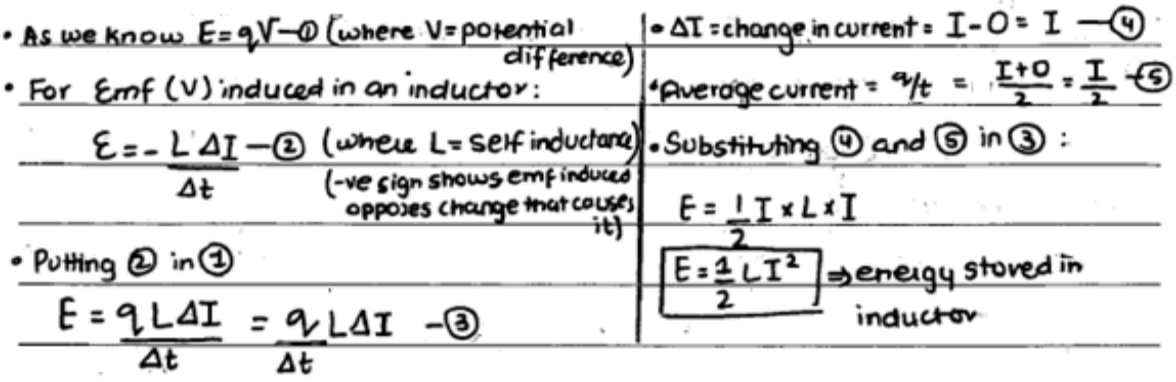
Any Additional Suggestions:

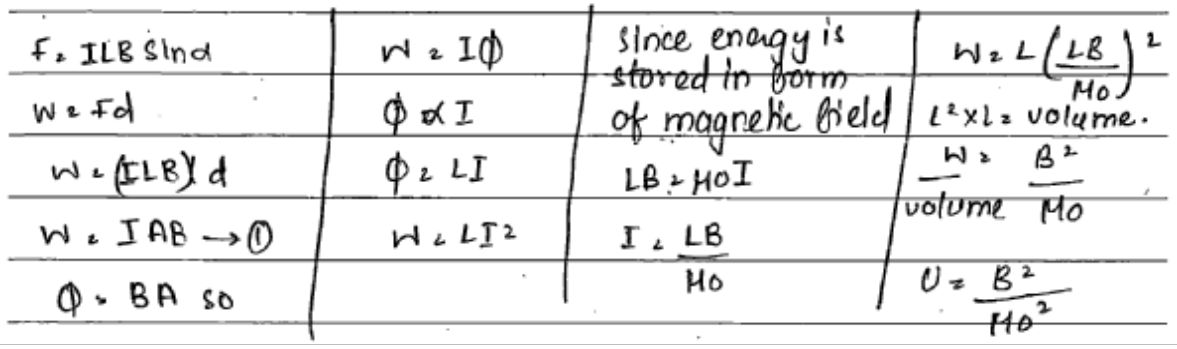
- Teachers should emphasise the connection between current-carrying conductors and moving charges by arranging and demonstrating the setup, and how the force on a conductor arises from individual charges.

- Teachers are encouraged to use online tools and simulations (e.g., PhET, Falstad) to visualise charge motion in magnetic fields, allowing students to adjust variables (q, v, *B*) and observe how the force changes, which allows students to interactively experiment with electric current and analyse its effects.

- Encouraging students to perform step-by-step derivations, while emphasising the importance of justifying and logically connecting each step, ensures that their methods are scientifically sound and improve the overall clarity and quality of their explanations.

- To address these gaps, targeted strategies are needed: concept mapping (compare ILB vs. qvB), step-validation drills (fix incomplete derivations), diagram integration annotate variables on diagrams), and pitfall workshops (analyse common errors). By strengthening these areas and diagram-based reasoning, students can develop more rigorous, complete solutions.

Question No. 3	
Question Text	Derive a relation for the energy stored in an inductor.
SLO No.	14.4.1
SLO Text	Describe the principle, construction and working of an AC and DC generator.
Max Marks	03
Cognitive Level	U
Checking Hints	1 mark for each highlighted step (3 required)
Overall Performance	Overall, the cohort failed to demonstrate competent understanding of energy storage in inductors, with few candidates correctly deriving the expression U = ½ L I² by analysing the relationship between induced emf, current growth, and work done.
Description of Better Responses	*Better responses* demonstrated clear proficiency in handling the question by logically developing the relationship for the energy stored in an inductor using fundamental concepts. They began with the expression for induced emf ($\varepsilon = L\frac{\Delta I}{\Delta t}$) and correctly related it to the work done in establishing current in the coil. Their step-by-step approach ultimately led to the correct final expression. These responses were characterised by well-organised work, accurate use of physical quantities, and clear explanations of how energy is stored in the magnetic field of the inductor. Such answers reflected not only a correct result but also a solid conceptual understanding supported by precise reasoning and structured presentation.
Image of Better Response	

Description of Weaker Responses	*Weaker responses* revealed several common challenges in deriving the energy relationship $U = \frac{1}{2} L I^2$. Many struggled with fundamental concepts, contained jumbled expressions, incorrectly stated the formula without justification or derived expressions that applied to capacitors $U = \frac{1}{2} C V^2$ instead of inductors. Some included irrelevant discussions about AC circuits or self-inductance definitions, indicating a lack of focus on the question's requirements. Some of the responses contained minor algebraic errors or incomplete explanations. A notable portion simply stated the final formula without demonstrating the crucial steps, revealing gaps in systematic problem-solving. Several responses confused the derivation with capacitor energy storage ($U = \frac{1}{2} C V^2$), while others diverged into unrelated topics like AC circuit applications, indicating the need for stronger conceptual differentiation. These results highlight class-wide comprehension of core principles but underscore the importance of reinforcing methodical derivations and clear distinction between similar electromagnetic concepts.
Image of Weaker Response	

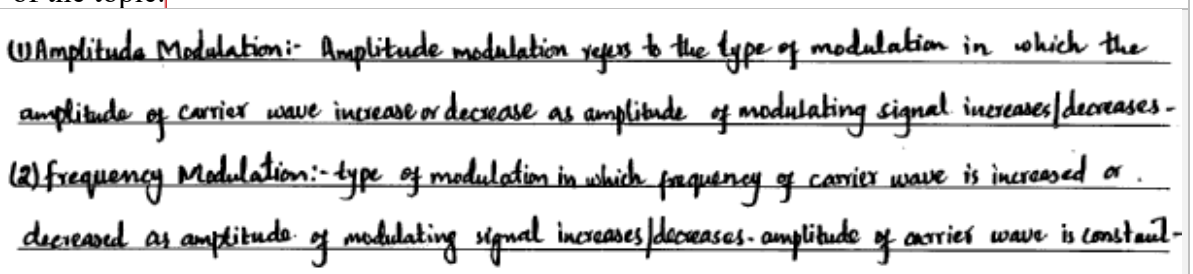
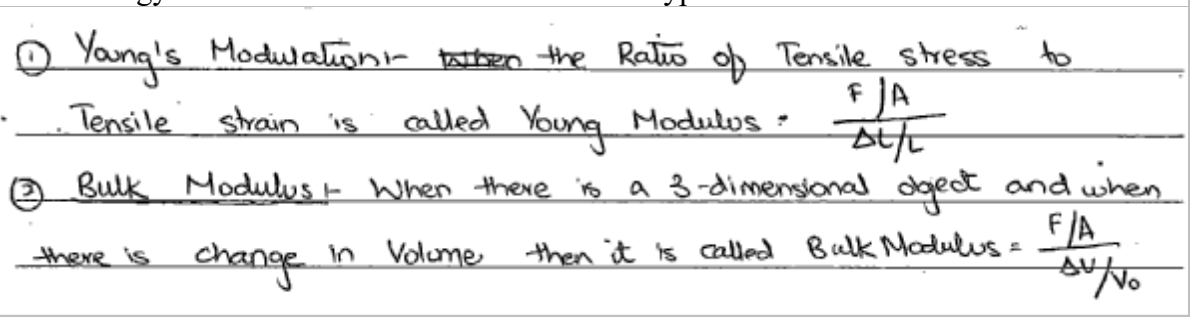
Suggestions for improvement (Highlight all that apply)

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Any Additional Suggestions:

- Teachers are encouraged to incorporate conceptual analogies by illustrating how the energy stored in an inductor depends on both the inductance and the square of the current. Emphasis should be placed on the origin of half of the factor as resulting from the gradual buildup of current and energy over time, which is represented graphically as the area under the current vs. time graph.

- Encourage students to perform step-by-step derivations while emphasising the importance of structure, clarity, and logical flow.

Question No. 4	
Question Text	State and explain TWO types of modulation.
SLO No.	15.53
SLO Text	Describe the amplitude modulation (AM) and frequency modulation (FM).
Max Marks	02
Cognitive Level	U
Checking Hints	1 mark for writing the correct statement in Amplitude Modulation 1 mark for writing the correct statement in Frequency Modulation
Overall Performance	The overall performance on this question was comparatively better, reflecting a clear understanding of the two types of modulation used in signal transmission. Some responses accurately defined both amplitude and frequency modulation with appropriate examples or brief explanations.
Description of Better Responses	*Better responses* provided clear definitions of both amplitude modulation (AM) and frequency modulation (FM). They correctly identified amplitude modulation as the process in which the amplitude of the carrier wave changes by the modulating signal, while the frequency remains constant. Frequency modulation was accurately described as the variation of the carrier wave's frequency based on the modulating signal, with the amplitude remaining constant. These responses were concise, technically accurate, and used appropriate terminology. Some candidates also strengthened their answers by including relevant examples or mentioning practical applications, reflecting a deeper understanding of the topic.
Image of Better Response	
Description of Weaker Responses	*Weaker responses* showed limited understanding of modulation types or provided incomplete definitions. Some responses defined only one type of modulation or mixed up the characteristics of amplitude and frequency modulation. Others used vague or incorrect terminology, failing to clearly state what aspect of the carrier wave is altered in each case. A few candidates confused the term "modulation" with unrelated concepts, such as modulus of elasticity, indicating a lack of conceptual clarity. These responses reflected gaps in foundational knowledge and highlighted the need for more focused instruction on key terminology and distinctions between modulation types.
Image of Weaker Response	

Suggestions for improvement (Highlight all that apply)

Maximising SLO Achievement	Pedagogy Used for that SLO	Assessment Strategies
• Identify the expectation of command words (use Command Word Guide) • Ensure the content is taught at the relevant cognitive level • Identify necessary content required (skills + concepts) • Review past paper questions on the concept • Utilise the resource guide for additional materials	• Story Board • Cause and Effect • Fish and Bone • Concept Mapping • Audio Visual Resources • Think, Pair and Share • Knowledge Platform videos • Questioning Technique (Socratic approach) • Practical Demonstration	• Past paper questions • Discussion on E-Marking Notes • AKU-EB Digital Learning Solution powered by Knowledge Platform https://akueb.knowledgeplatform.com/login

Any Additional Suggestions:

- Teachers are suggested to incorporate visual aids and real-world examples (such as AM/ FM radio) to help students connect theory with practical applications and strengthen conceptual clarity.
- Students should be encouraged to practice writing complete, structured responses, including both definitions and distinguishing features, and to revise their understanding through peer discussion or collaborative activities.

Question No. 5	
Question Text	Differentiate between elastic and plastic deformations in solids by giving any TWO points.
SLO No.	16.2.1
SLO Text	Differentiate between elastic and plastic deformations in solids.
Max Marks	02
Cognitive Level	U
Checking Hints	1 mark for writing each difference (Any 2 required)
Overall Performance	The overall performance of the cohort on this question was good. Many candidates correctly identified the key distinctions between elastic and plastic deformation, and many effectively supported their responses by making and drawing graphical representations. These answers reflected a clear understanding and familiarity with the concepts.
Description of Better Responses	*Better responses* clearly and accurately stated two well-defined differences between elastic and plastic deformation. They correctly described elastic deformation as temporary and reversible, where the material returns to its original shape after the removal of force and plastic deformation as permanent, resulting in a lasting change in shape. Additionally, they noted that elastic deformation occurs within the proportional limit, following Hooke's Law, while plastic deformation begins beyond this limit, where stress and strain are no longer proportional. These responses often included stress-strain graphs to visually support their explanations, clearly distinguishing the linear region representing elastic behaviour from the non-linear region associated with plastic deformation. The use of precise scientific language and correct terminology reflected strong conceptual understanding and effective communication of key physical principles.

Image of Better Response

S.No.	Elastic Deformations	Plastic Deformations
1	In elastic deformaties, when stress is removed the object returns to its original shape.	In plastic deformaties, when stress is removed the object is permanently deformed, it can't return to its original shape.
2	It obey's Hooke's Law, as the graph between stress and strain is linear (straight line).	It does not obey hooke's Law as the graph between stress and strain is non-linear.

Description of Weaker Responses

Weaker responses showed limited understanding of the key distinctions between elastic and plastic deformation. Some responses mentioned only one difference or repeated the same idea in different words, failing to provide a second valid point. Others used vague or incorrect language, such as referring to the deformations as "soft" or "hard," without explaining the concepts of reversibility or permanence. A few responses confused the topic entirely, providing unrelated definitions or mixing it up with other mechanical properties like elasticity or stiffness. These responses lacked clarity, accuracy, and proper use of scientific terminology, indicating a need for more focused instruction on the fundamental concepts. Students should be encouraged to use accurate scientific terms such as "reversible", "permanent", "Hooke's Law", and "proportional limit" when describing deformation types, avoiding vague or informal language.

Image of Weaker Response

S.No.	Elastic Deformations	Plastic Deformations
1	Elastic properties are easy to deform than plastic.	Plastic property is hard to deform than elastic.
2	Stretching properties present.	Stretching properties not present or unstretchable

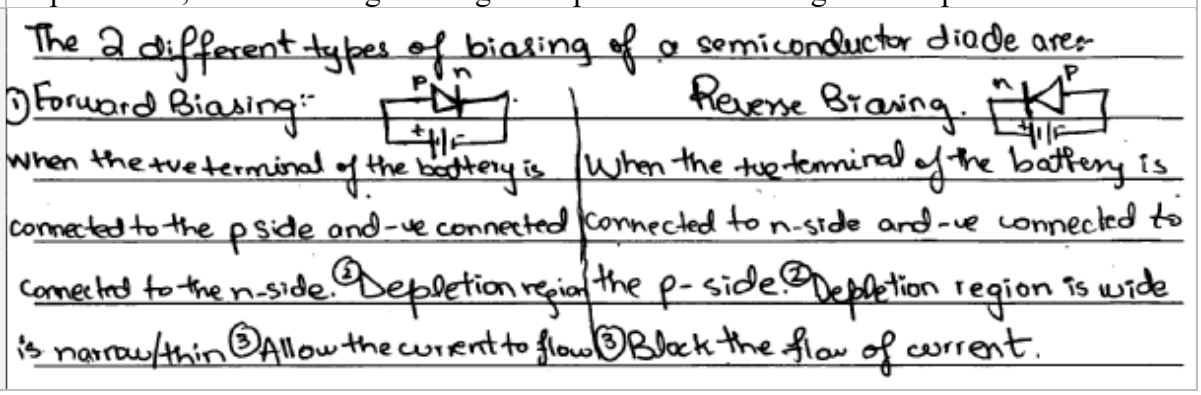
Suggestions for improvement (Highlight all that apply)

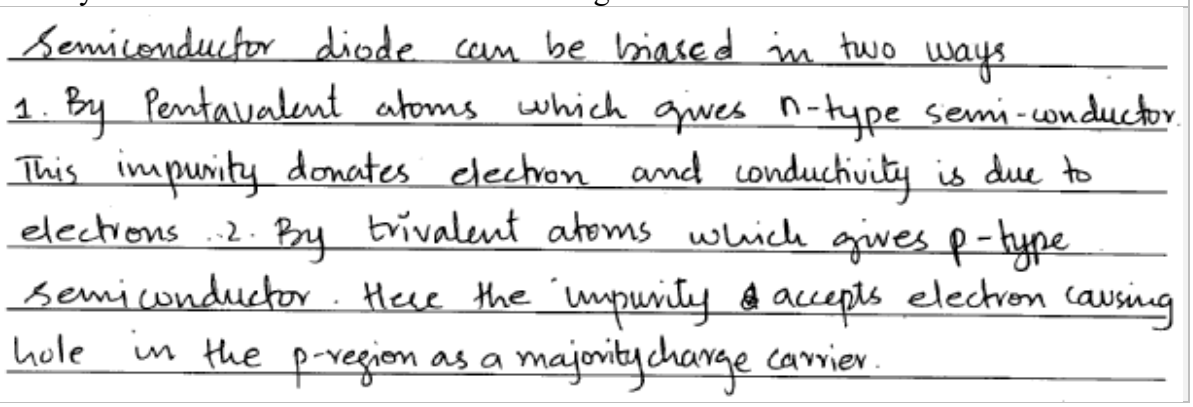
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Any Additional Suggestions:

- Teachers should emphasise the conceptual differences using visual tools, such as stress-strain graphs, clearly highlighting the linear (elastic) and non-linear (plastic) regions, as well as the proportional limit and yield point.

- Teachers are encouraged to provide structured comparison exercises, asking students to write side-by-side differences between similar physical concepts (e.g., elastic vs. plastic, stress vs. pressure) to strengthen analytical thinking.

Question No. 6	
Question Text	Describe TWO different types of biasing of a semiconductor diode.
SLO No.	17.2.5
SLO Text	Describe forward and reverse bias.
Max Marks	03
Cognitive Level	U
Checking Hints	1 mark for writing the CORRECT name of biasing 1 mark for describing EACH biasing (Any 2 required)
Overall Performance	Overall, the performance on this question was not satisfactory, with a majority of candidates unable to demonstrate a sound understanding of the two main biasing techniques. Only some of the responses accurately described forward and reverse biasing, including the behaviour of the diode under each condition.
Description of Better Responses	*Better responses* clearly described both forward and reverse biasing techniques of a *p-n* junction diode with accurate explanations of the diode's behaviour under each condition. They correctly explained that in forward biasing, the positive terminal of the battery is connected to the *p*-side and the negative to the *n*-side, reducing the depletion region and allowing current to flow. In contrast, reverse biasing was described as connecting the positive terminal to the n-side and the negative to the p-side, which widens the depletion region and prevents current flow (except for a small leakage current). These responses used appropriate terminology and often included labelled diagrams to visually support their explanations, demonstrating a strong conceptual understanding of the topic.
Image of Better Response	

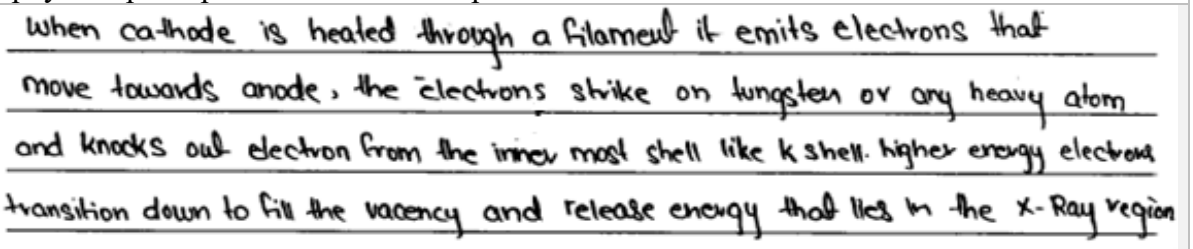
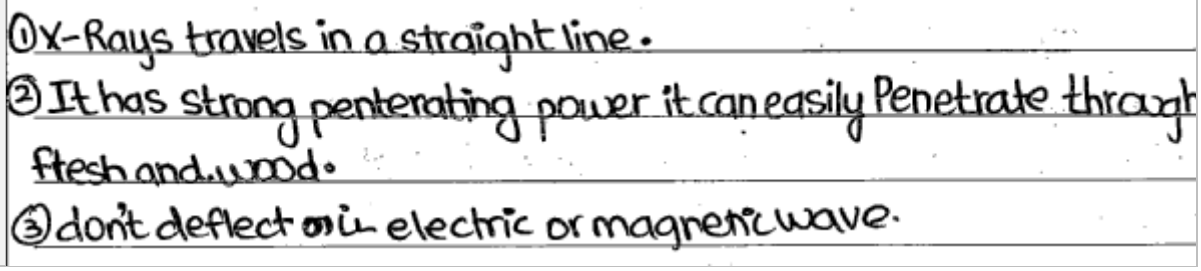
Description of Weaker Responses	*Weaker responses* demonstrated limited understanding of the biasing techniques of a *p-n* junction diode or failed to provide complete and accurate descriptions. Some responses merely stated the terminal connections without explaining the resulting behaviour of the diode, such as changes in the depletion region or current flow under forward or reverse bias. Others incorrectly interchanged the two biasing conditions, showing confusion about when the diode conducts or blocks current. A few responses were off-topic, discussing rectification processes like half-wave or full-wave rectification, or unrelated concepts such as doping techniques, instead of the required biasing methods. These responses lacked conceptual clarity, appropriate terminology, and often omitted key physical explanations, highlighting the need for more focused instruction and targeted practice in this area. However, several candidates focused only on terminal connections without explaining how the diode conducts or blocks current in each case. Some responses showed confusion between forward and reverse biasing, and a few candidates digressed by discussing rectification circuits instead of addressing the biasing techniques directly. This indicates that while the basic concept is understood by many, further reinforcement is needed to clarify diode behaviour under different biasing conditions.
Image of Weaker Response	

Suggestions for improvement (Highlight all that apply)

Maximising SLO Achievement	Pedagogy Used for that SLO	Assessment Strategies
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Any Additional Suggestions:

- Teachers should emphasise not only the terminal connections but also the physical behaviour of the diode under each biasing condition, such as changes in the depletion region, current flow, and barrier potential, using clear illustrations and circuit diagrams.
- Teachers are encouraged to use visual aids like animated simulations or labelled diagrams to help students visualise diode behaviour under different biasing conditions, reinforcing conceptual clarity.

Question No. 7	
Question Text	In the process of inner shell transitions, explain the emission of the characteristics of X-rays.
SLO No.	19.4.1
SLO Text	Describe inner shell transitions.
Max Marks	02
Cognitive Level	U
Checking Hints	1 mark for writing each point (2 required) (**Note**: 1 mark will be given if only the characteristics of X-rays are mentioned)
Overall Performance	The overall performance on this question was very weak. Many responses incorrectly described that characteristic X-rays are produced when high-energy electrons knock out inner-shell electrons of a target atom, and the resulting vacancy is filled by an electron from a higher energy level, releasing energy in the form of X-ray photons.
Description of Better Responses	*Better responses* demonstrated a thorough understanding of characteristic X-ray production via inner-shell transitions. They correctly described that when high-energy electrons strike a metal target, they can eject electrons from the inner shell (typically the K-shell) of the atom. This creates a vacancy, which is then filled by an electron from a higher energy level (such as the L or M shell). The energy difference between these two levels is released in the form of a photon and this photon corresponds to a characteristic X-ray whose energy is specific to the target material. Many of these responses also highlighted the basic requirements for X-ray production, including a high-voltage source to accelerate electrons and a metal target with a high atomic number. Some candidates supported their explanations with labelled diagrams, helping to illustrate the shell transition clearly. These responses reflected strong conceptual understanding, accurate terminology, and the ability to connect physical principles with observable phenomena.
Image of Better Response	
Description of Weaker Responses	*Weaker responses* demonstrated a limited understanding of the process involved in the production of characteristic X-rays. Some responses lacked reference to inner-shell transitions entirely or only mentioned high-energy electrons without explaining how they interact with the target material. Others described the ejection of electrons but failed to mention the subsequent transition of electrons from higher energy levels or the emission of X-ray photons. Some candidates confused the concept with general properties of X-rays or incorrectly described phenomena such as electron excitation and photon absorption, indicating a mix-up with atomic absorption rather than emission. These responses were often vague, lacked key technical terms, and showed gaps in understanding the sequence of events required for characteristic X-ray production. A notable portion of responses were incomplete or inaccurate. Some omitted critical steps (e.g., electron ejection or specific transitions), while others confused the mechanism with electron excitation/ photon absorption or digressed into general X-ray properties.
Image of Weaker Response	

Suggestions for improvement (Highlight all that apply)

Maximising SLO Achievement	Pedagogy Used for that SLO	Assessment Strategies
• Identify the expectation of command words (use Command Word Guide) • Ensure the content is taught at the relevant cognitive level • Identify necessary content required (skills + concepts) • Review past paper questions on the concept • Utilise the resource guide for additional materials	• Story Board • Cause and Effect • Fish and Bone • Concept Mapping • Audio Visual Resources • Think, Pair and Share • Knowledge Platform videos • Questioning Technique (Socratic approach) • Practical Demonstration	• Past paper questions • Discussion on E-Marking Notes • AKU-EB Digital Learning Solution powered by Knowledge Platform https://akueb.knowledgeplatform.com/login

Any Additional Suggestions:

- Teachers should emphasise the sequential process of characteristic X-ray production, starting from electron acceleration, inner-shell ionisation, and energy release due to electronic transitions. Use step-by-step diagrams to visually reinforce each stage.

- Students should be encouraged to distinguish clearly between characteristic and continuous X-rays, focusing on the role of inner-shell transitions and the dependence of characteristic X-rays on the atomic structure of the target material.

Question No. 8	
Question Text	Describe, in three steps, how a nuclear reactor generates electricity.
SLO No.	20.9.1
SLO Text	Explain the working principle of a nuclear reactor.
Max Marks	03
Cognitive Level	K
Checking Hints	1 mark for writing each point (3 required)
Overall Performance	Overall, candidates addressed this question not very effectively; they were unable to demonstrate a sound understanding of how electricity is generated in a nuclear reactor. Few responses correctly outlined the key steps involved, including nuclear fission in the reactor core, heat generation, the use of this heat to produce steam, steam-driven turbine rotation, and electricity generation through a connected generator. Some candidates strengthened their responses by including block diagrams and clearly explaining the function of each component, such as the control rods, moderator, coolant, and heat exchanger.

Description of Better Responses	*Better responses* provided a clear, step-by-step explanation of how electricity is generated in a nuclear reactor, demonstrating both accuracy and depth of understanding. These responses began with the process of nuclear fission occurring in the reactor core, where heavy nuclei such as uranium-235 split to release a large amount of thermal energy. Candidates also correctly explained the roles of the moderator, which slows down neutrons to sustain the chain reaction, and the control rods, which absorb excess neutrons to regulate the rate of fission. They then accurately described how the heat produced is transferred via a coolant to a heat exchanger, where it converts water into steam. The steam was correctly identified as driving a turbine, which in turn spins a generator to produce electricity. Some responses also included well-labelled block diagrams that illustrated the flow of energy and identified the roles of key components. The use of correct terminology and logical sequencing reflected a solid grasp of the nuclear power generation process.
Image of Better Response	A nuclear reactor consists of 4 main components; 1. Core: provides the fuel like $^{235}_{92}U$ for the initiation of fission reaction. 2. Moderator: substance (like heavy water) that slows down the neutrons and favours fission reaction. 3. Control rods: rods (like boron rods) control the speed of reaction by absorbing neutrons. 4. Heat exchanger: uses heat energy to produce steam which moves the turbine and generates electricity.
Description of Weaker Responses	*Weaker responses* demonstrated a limited understanding of the sequential process involved in generating electricity through a nuclear reactor. Some responses lacked structure or skipped key steps, such as the conversion of thermal energy to steam or the role of the turbine and generator. Others focused too narrowly on the concept of nuclear fission without explaining how the energy released is harnessed to produce electricity. A few candidates confused nuclear power generation with unrelated processes, referencing chemical energy or chemical reactions, while others incorrectly described nuclear fusion instead of fission. Additionally, several responses did not identify or explain the function of critical components such as the moderator, control rods, coolant, or heat exchanger. These responses reflected gaps in conceptual clarity and indicated a need for more focused instruction on the complete energy conversion pathway within a nuclear reactor. Some responses lacked detail or skipped steps, resulting in incomplete explanations. A few candidates confused the concept by focusing on nuclear fusion or erroneously referencing chemical energy.
Image of Weaker Response	nuclear reactor generate electricity using chemica of different- nuclear power and a structure which processes and converst the nuclear energy to electrical energy. using mechanisms. and nuclear power is the source. for generating and converting energy in different forms.

Suggestions for improvement (Highlight all that apply)

Maximising SLO Achievement	Pedagogy Used for that SLO	Assessment Strategies
• Identify the expectation of command words (use Command Word Guide) • Ensure the content is taught at the relevant cognitive level • Identify necessary content required (skills + concepts) • Review past paper questions on the concept • Utilise the resource guide for additional materials	• Story Board • Cause and Effect • Fish and Bone • Concept Mapping • Audio Visual Resources • Think, Pair and Share • Knowledge Platform videos • Questioning Technique (Socratic approach) • Practical Demonstration	• Past paper questions • Discussion on E-Marking Notes • AKU-EB Digital Learning Solution powered by Knowledge Platform https://akueb.knowledgeplatform.com/login

Any Additional Suggestions:

- Teachers should emphasise the sequential flow of energy conversion from nuclear fission to thermal energy, steam production, turbine rotation, and finally, electricity generation using labelled flowcharts or block diagrams to reinforce the stages.

- Students should be encouraged to understand the function of each key component (reactor core, control rods, moderator, coolant, heat exchanger, turbine, generator), not just memorise their names. Role-based explanations help build deeper conceptual understanding.

Extended Response Questions (ERQs)

Extended response questions offered a choice between parts ‘a’ and ‘b’

Question No. 9a	
Question Text	i. Make a circuit diagram of the Wheatstone bridge. ii. Derive the relationship between the resistances in a Wheatstone bridge circuit.
SLO No.	12.3.6
SLO Text	Calculate the value of carbon resistance by using colour codes.
Max Marks	07
Cognitive Level	A
Checking Hints	i. 3 marks for the circuit diagram (1 mark for correctly mentioned resistors, 1 for the Galvanometer, and 1 for the power source) 1 mark for the correct direction of current in the circuit ii. 1 mark for writing the mathematical step in the derivation (3 required)

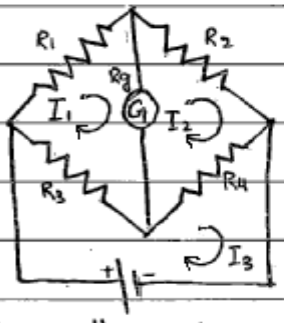
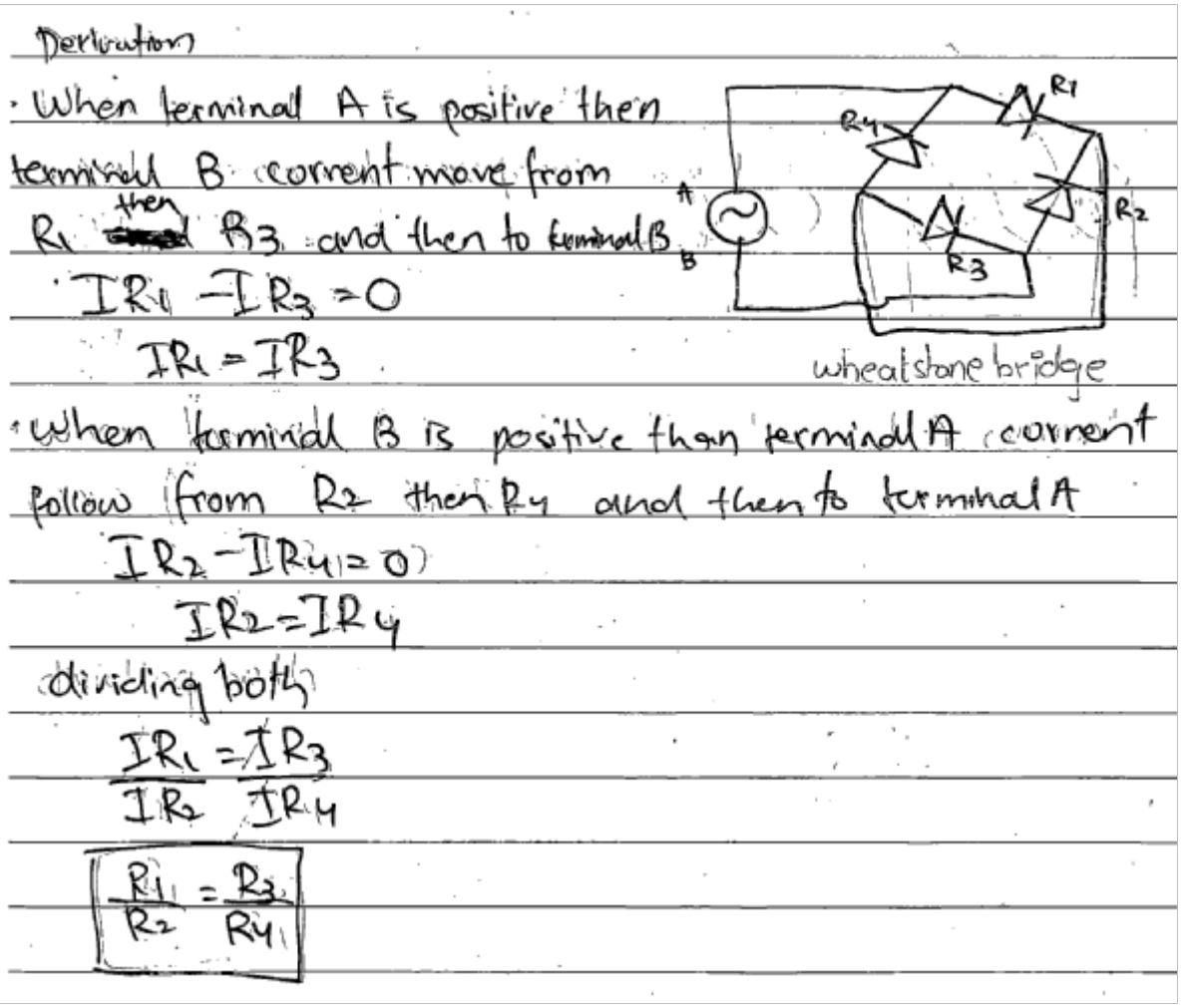
Overall Performance	Candidates opted for this question comparatively more than part B. Amongst the ones who opted for this question, the performance was satisfactory. A few of the candidates successfully attempted this part of the question and drew the correct and clearly labelled Wheatstone bridge circuit, indicating the direction of current, and accurately derived the relationship between the resistors under the balanced bridge condition using Kirchhoff's voltage law.
Description of Better Responses	*Better responses* accurately addressed both parts of the question with clarity and precision. They correctly drew a well-labelled Wheatstone bridge circuit, clearly identifying all four resistors, the galvanometer between the midpoints, and the power supply across the outer nodes. The direction of the current was appropriately indicated, showing an understanding of current flow in the circuit. In the second part, these candidates applied Kirchhoff's voltage law effectively to the two loops involving the galvanometer and derived the condition for bridge balance. They logically arrived at the relationship $\frac{R_1}{R_2} = \frac{R_3}{R_4}$, clearly showing each step with proper justification. The use of correct terminology, systematic working, and neat presentation reflected strong conceptual understanding and confident application of circuit analysis principles.
Image of Better Response	a (i)  Dividing current into three loops I_1, I_2, I_3 (ii) Using Kirchoffs voltage law $-I_1R_1 - (I_1-I_3)R_3 - (I_1-I_2)R_g = 0$ —① $-I_2R_2 - (I_2-I_3)R_4 - (I_2-I_1)R_g = 0$ —② Now assuming galvanometer shows 0 deflection $I_1 = I_2 = I$ $-IR_1 - (I-I_3)R_3 - (I-I)R_g = 0$ —③ $-IR_2 - (I-I_3)R_4 - (I-I)R_g = 0$ —④ Dividing eq 3 and 4 $\frac{IR_1}{IR_2} = \frac{(I-I_3)R_3}{(I-I_3)R_4} \rightarrow \boxed{\frac{R_1}{R_2} = \frac{R_3}{R_4}}$ → Answer (Final relation)
Description of Weaker Responses	*Weaker responses* showed limited understanding of the Wheatstone bridge circuit and its balance condition. Some responses included incomplete or incorrectly drawn circuit diagrams, such as misplacing the galvanometer, omitting labels, or incorrectly connecting the power supply. Others failed to indicate the direction of current, suggesting a lack of awareness of circuit flow. In the derivation part, many candidates did not apply Kirchhoff's voltage law correctly or skipped essential steps, resulting in an incorrect or unsupported relationship between the resistors. A few responses demonstrated significant confusion by drawing the diagram of a bridge rectifier instead of a Wheatstone bridge, indicating a misunderstanding of the question. Overall, these responses reflected gaps in both conceptual clarity and application of basic circuit analysis techniques. A few responses only attempted the diagram or left the derivation incomplete. Additionally, some candidates misunderstood the question entirely and incorrectly drew the circuit diagram of a bridge rectifier, indicating confusion between unrelated circuit concepts. This reflects the need for clearer conceptual distinction and targeted practice on the application of Kirchhoff's laws in bridge circuits.

Image of Weaker Response



Suggestions for improvement (Highlight all that apply)

Maximising SLO Achievement	Preferred Pedagogy Used for this SLO	Assessment Strategies
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Any Additional Suggestions:

- Teachers should emphasise the structural components and layout of the Wheatstone bridge, ensuring students can distinguish it clearly from other circuits like the bridge rectifier. Repeated diagram-based practice with different resistor values and configurations can reinforce visual recognition.

- Teachers are encouraged to provide more guided practice with Kirchhoff's laws and circuit analysis, especially in multi-loop or bridge circuits, which can significantly strengthen candidates' conceptual understanding and problem-solving skills.

Question No. 9b	
Question Text	A cell of electromotive force (e.m.f.) 2.0 V provides power to an external circuit consisting of two resistors of 12 Ω each, which are joined in parallel combination in a circuit and a voltmeter which reads a voltage of 1.6 V. i. Make a circuit diagram of the given situation. ii. Calculate the internal resistance of the cell and the power dissipated by the internal resistance.
SLO No.	12.4.6
SLO Text	Calculate the power dissipation due to the internal resistance of a circuit.
Max Marks	07
Cognitive Level	A
Checking Hints	i. 4 marks for circuit diagram (2 for resistors, 1 for battery and 1 for voltmeter) ii. 1 mark for the mathematical steps (3 required)
Overall Performance	Overall, candidates addressed this question with very low success; several recurring misconceptions were observed. Many candidates were unable to draw a reasonably accurate and labelled circuit diagram showing two 12 Ω resistors connected in parallel to a 2V battery with internal resistance, and a voltmeter measuring the terminal voltage. These responses incorrectly interpreted the voltage drop to set up calculations for internal resistance and power dissipation.
Description of Better Responses	In *better responses*, candidates accurately constructed a labelled circuit diagram showing the correct configuration of two 12 Ω resistors connected in parallel across a cell with internal resistance and a voltmeter across the load. They correctly identified the effective resistance of the parallel combination as 6 Ω and used Ohm's law to calculate the total current in the circuit from the terminal voltage (1.6 V). These responses showed clear application of the formula V = I R to determine current, followed by using V = ε m f – I r to calculate internal resistance. In the final step, candidates successfully calculated the power dissipated in the internal resistance using the expression $P = I^2 r$, presenting their work neatly and logically. The responses demonstrated a strong grasp of electrical circuit analysis, proper use of physical laws and units and clear communication of reasoning through well-organised steps and appropriate labelling.

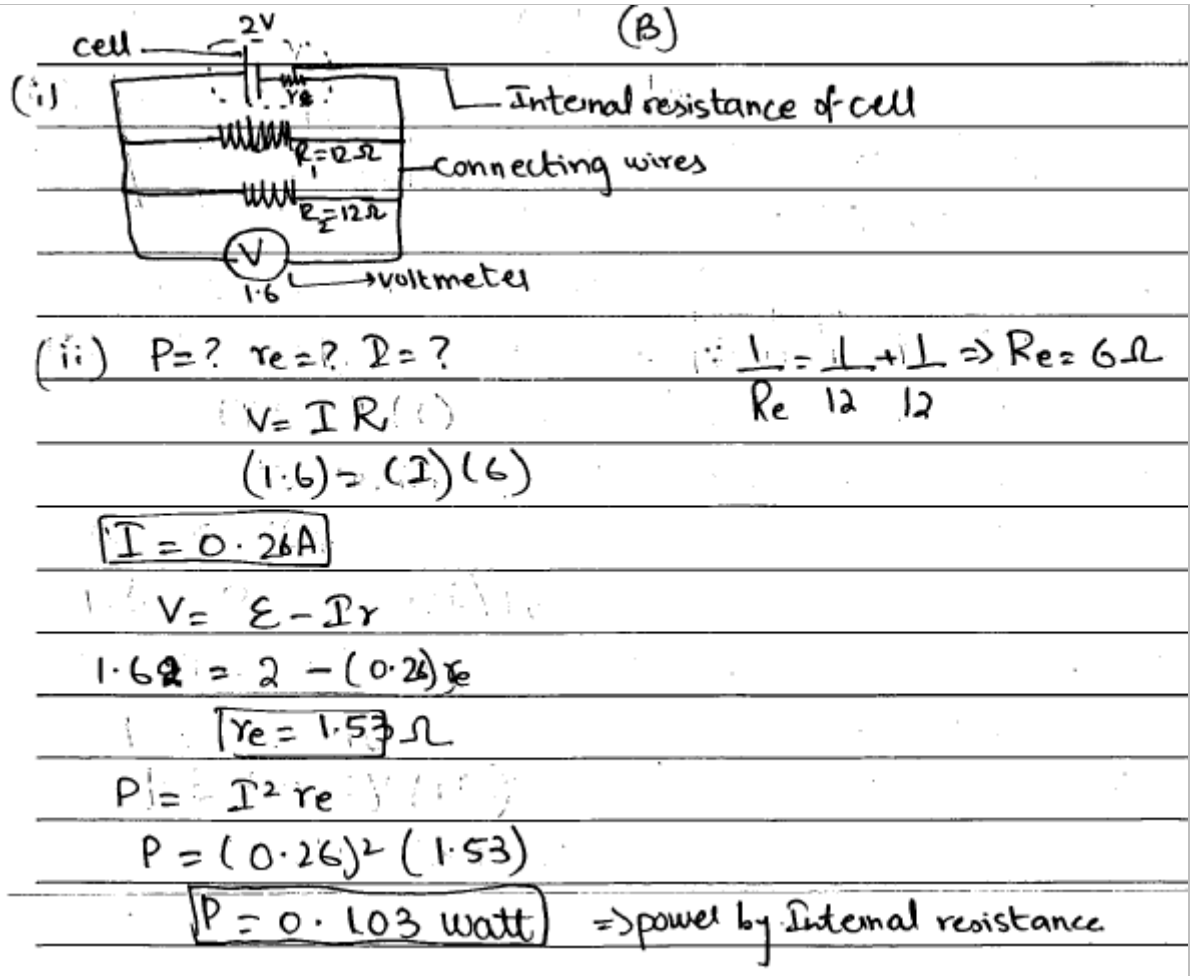
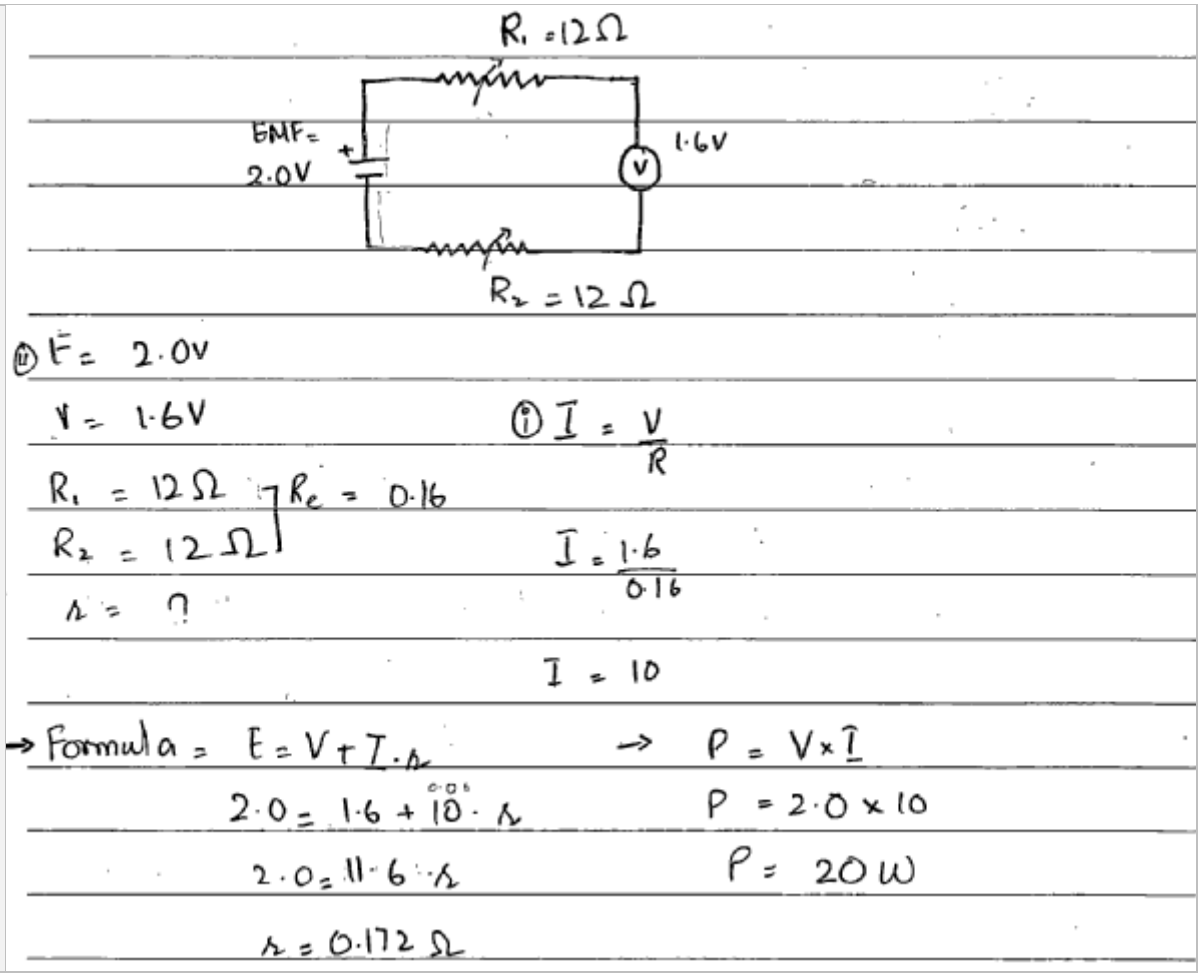
Image of Better Response	
Description of Weaker Responses	*Weaker responses* demonstrated limited understanding of the parallel circuit configuration and the concept of internal resistance. Some responses incorrectly drew the resistors in series rather than parallel or omitted key components such as the voltmeter or internal resistance in the circuit diagram. Others showed confusion between terminal voltage and emf, often misapplying Ohm's law or using inconsistent values in the calculations. Several candidates attempted to use formulas without proper justification or skipped essential steps, resulting in incorrect or unsupported answers for internal resistance or power. Additionally, some responses lacked proper labelling, unit usage, or logical flow, making it difficult to follow their reasoning. These issues reflect gaps in understanding of basic circuit principles and highlight the need for more structured practice with labelled diagrams and step-by-step problem-solving strategies. These varied responses indicate that while the basic concepts of e.m.f., terminal voltage, and internal resistance are understood by many, there is a need for further reinforcement in accurately applying them within circuit diagrams and in performing related calculations.

Image of Weaker Response	

Suggestions for improvement (Highlight all that apply)

Maximising SLO Achievement	Preferred Pedagogy Used for this SLO	Assessment Strategies
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Any Additional Suggestions:

- Teachers are encouraged to provide targeted practice on interpreting and constructing circuit diagrams, with special focus on distinguishing between series and parallel configurations and correctly placing components like voltmeters, ammeters, and internal resistances.
- Students should be encouraged to relate each physical quantity (emf, terminal voltage, current, resistance, and power) to the relevant equation and understand the context in which it applies. Practice deriving formulas from first principles can deepen conceptual clarity.

Question No. 10a	
Question Text	Explain why the classical theory of light fails to account for the: i. concept of threshold frequency, ii. direct proportionality between the intensity of incident light and the photoelectric current.
SLO No.	18.3.2
SLO Text	Explain different features of the photoelectric effect using a graph.
Max Marks	07
Cognitive Level	U
Checking Hints	1 mark for each point (7 required)
Overall Performance	Few candidates opted for part A. The ones who attempted this part of the question failed to address its requirements, demonstrating a weak grasp of the photoelectric effect and the limitations of classical physics in explaining key observations. A few responses accurately pointed out that classical wave theory fails to account for the existence of a threshold frequency. It predicts that electron emission should occur regardless of the light's frequency if the intensity is sufficient, which contradicts experimental evidence. Similarly, candidates explained that classical theory could not predict the immediate emission of electrons or their dependence on the frequency of light rather than its intensity. Some candidates strengthened their answers with well-labelled graphs illustrating the relationship between photocurrent and intensity, as well as kinetic energy versus frequency.
Description of Better Responses	*Better responses* demonstrated a strong conceptual understanding of the shortcomings of classical physics in explaining the photoelectric effect. They clearly explained that, according to classical wave theory, increasing the intensity of light, regardless of frequency, should increase the energy transferred to electrons and eventually lead to their emission. However, they contrasted this with experimental observations, which show that no electrons are emitted if the incident light is below a certain threshold frequency, even at high intensity. These candidates also correctly stated that classical physics could not explain the immediate emission of electrons when light above the threshold frequency is used. In addition, many candidates supported their explanations with well-drawn and labelled graphs, such as the graph of photoelectric current versus intensity, and kinetic energy of emitted electrons versus frequency. The use of appropriate scientific terms, including "threshold frequency", "instantaneous emission" and "quantization of energy", enhanced the clarity and technical accuracy of their responses. Some candidates also referred to Einstein's photon model to explain the quantum perspective, further strengthening their answers. These responses reflected both clarity and depth, showing a well-rounded understanding of why classical theory fails when compared to experimental results.

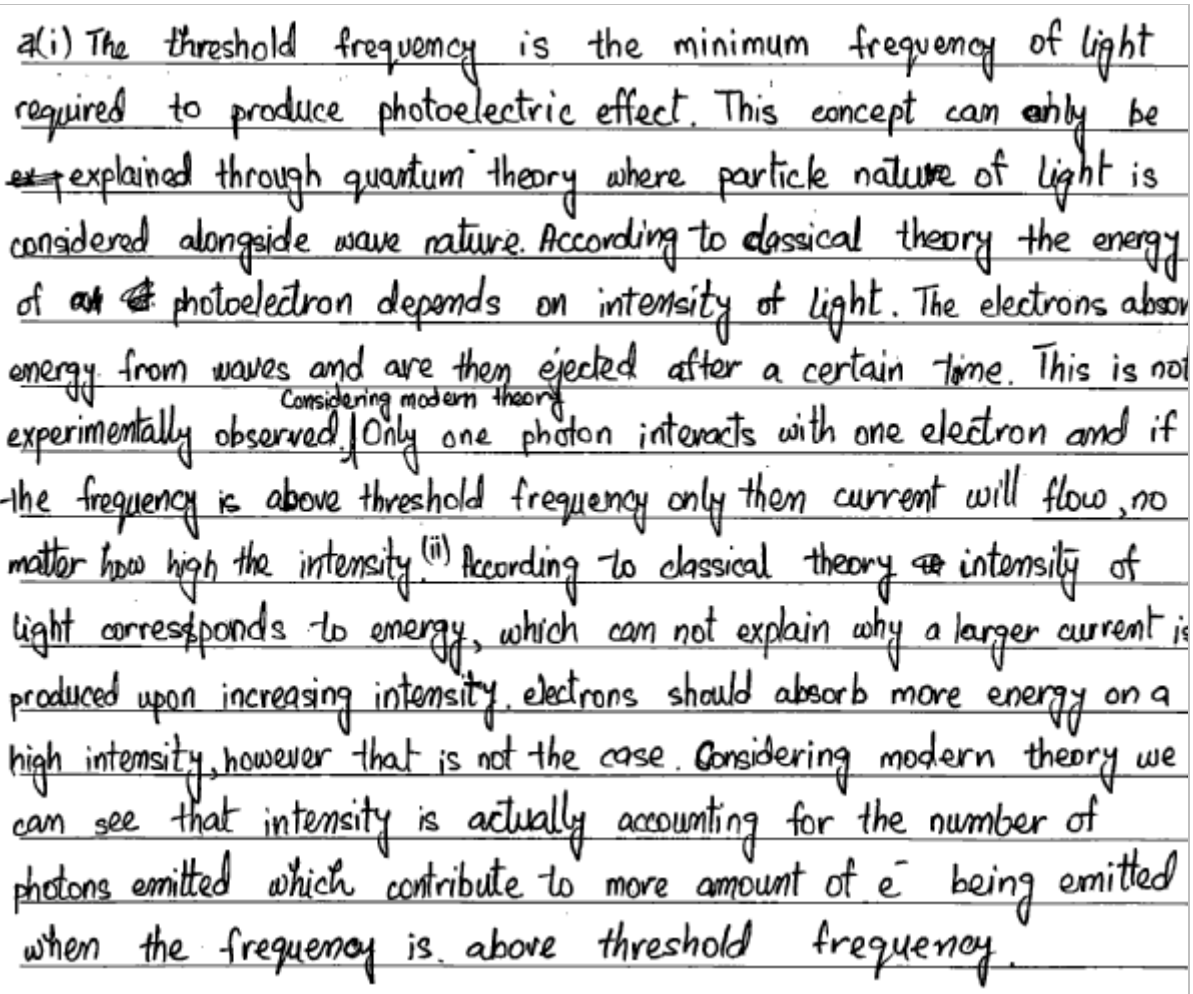
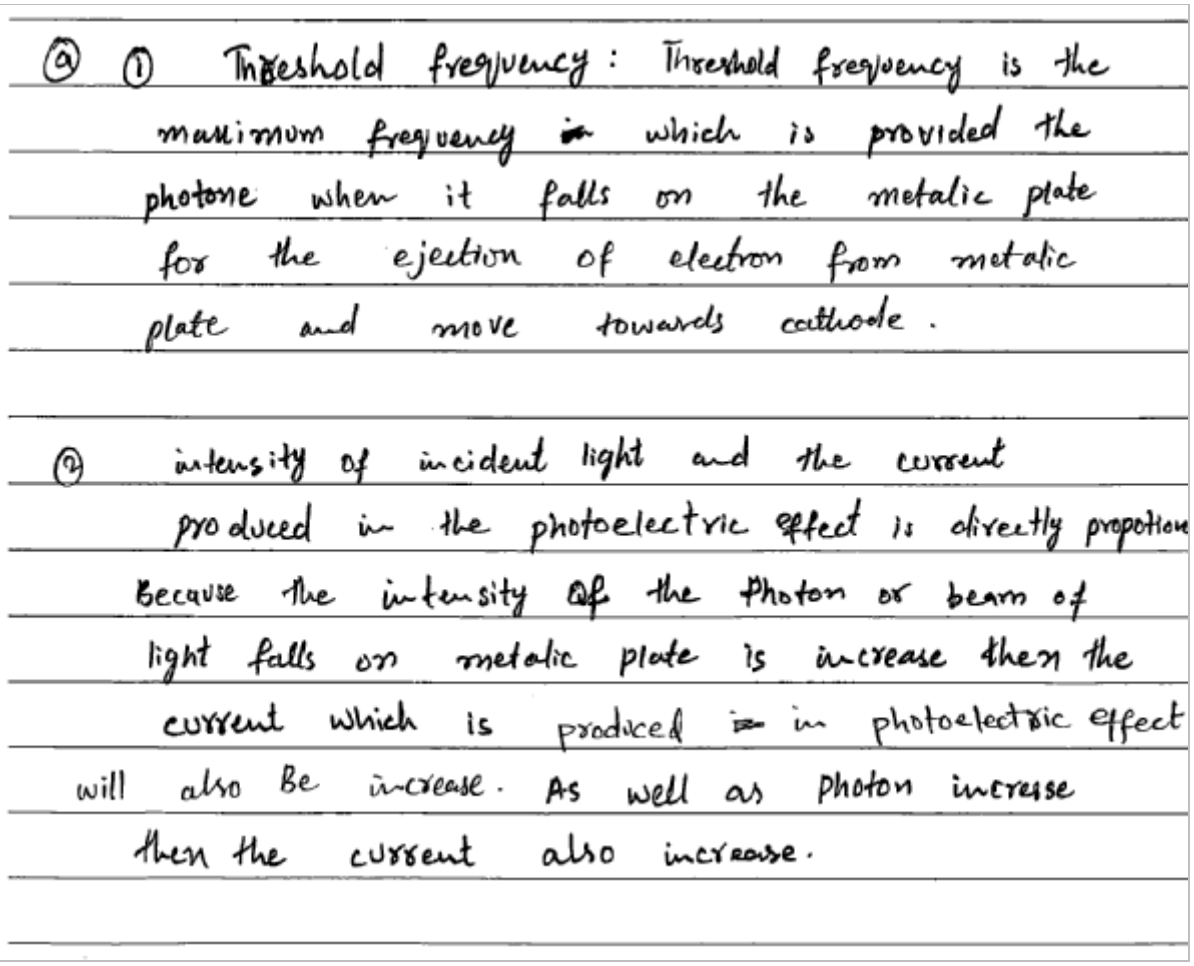
Image of Better Response	
Description of Weaker Responses	*Weaker responses* showed limited understanding of the specific ways in which classical physics fails to explain key features of the photoelectric effect. Some responses simply described the photoelectric effect in general terms without addressing the question's focus on threshold frequency and the relationship between light intensity and photocurrent. Others confused or incorrectly explained the concepts, such as stating that intensity affects the energy of emitted electrons, which reflects a misunderstanding of the distinction between frequency and intensity. A few responses misused terminology or lacked logical structure, making it difficult to follow the argument. Additionally, there were instances where candidates relied solely on memorised definitions without applying them to explain the failure of classical theory. These responses highlighted gaps in conceptual clarity and suggested the need for more focused instruction on the differences between classical and quantum models of light.

Image of Weaker Response	

Suggestions for improvement (Highlight all that apply)

Maximising SLO Achievement	**Pedagogy Used for that SLO**	**Assessment Strategies**
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Any Additional Suggestions:

Nil

Question No. 10b	
Question Text	i. Define inertial and non-inertial frame of reference. ii. Show that the Earth can be viewed both as an inertial and non-inertial frame of reference.
SLO No.	18.4.2
SLO Text	Compare the phenomenon of pair production and pair annihilation.
Max Marks	07
Cognitive Level	U
Checking Hints	1 mark for each definition (2 required) 1 mark for writing each point on Earth as an Inertial Frame (2 required) 1 mark for writing each point on Earth as an Inertial Frame (3 required)
Overall Performance	Though most of the candidates opted for this question, but the overall performance was not well. The candidates were unable to demonstrate a clear understanding of the key concepts. A few candidates correctly defined inertial and non-inertial frames of reference and provided relevant justifications for how Earth can be considered as both, depending on the context. Some responses included multiple points and examples, such as Earth's slow rotation being negligible in everyday situations (inertial) and the need to consider fictitious forces in more precise or large-scale systems (non-inertial).
Description of Better Responses	*Better responses* accurately defined both inertial and non-inertial frames of reference and provided well-reasoned justifications supported by appropriate examples. They correctly identified an inertial frame as one in which Newton's laws hold without the need for pseudo forces, typically involving either rest or motion with uniform velocity. Non-inertial frames were described as accelerating frames where Newton's laws require modification through the inclusion of fictitious forces. Candidates also clearly explained how Earth can be viewed from both perspectives. They justified that Earth, although rotating and revolving (and therefore technically accelerating), can be approximated as an inertial frame for most practical purposes due to the relatively small magnitude of its centripetal acceleration. Some candidates supported their justification by referring to the Earth's vast radius and the negligible effects of this acceleration on local, everyday observations. Others mentioned that in more sensitive cases, such as large-scale geophysical phenomena or weather systems, Earth's rotation introduces measurable non-inertial effects like the Coriolis force. The use of appropriate terminology, accurate reasoning, and structured explanations reflected strong conceptual understanding and effective application of physical principles.

Images of Better Responses	(i) • Inertial Frame : A frame of reference in which coordinate system obey's law of inertia (newton's 1st law of motion) · furthermore it is a non-accelerating frame either at rest or move with uniform velocity · • Non-Inertial Frame : A frame of reference in which coordinate system does not obey law of inertia (newton's 1st law) · Furthermore it is a accelerating frame moving with variable velocity · (ii) Earth is considered as an approximate inertial frame however it is a non-inertial frame · Earth rotates around the sun in circular path even if its velocity (magnitude) is constant, it's direction keep on changing constantly due to centripetal acceleration (a_c) acting on it . Thus it is considered e that Earth is accelerating so it is a ~~non-accelerating~~ non-inertial frame · However the effect of centripetal acceleration (a_c) of Earth is very small due to its huge circular path that it is not feel like accelerating · In this regard Earth is usually considered as an approximate inertial frame ·
Description of Weaker Responses	In *weaker responses*, candidates had a limited understanding of the key concepts or failed to fully address the question. Some responses provided only brief or vague definitions of inertial and non-inertial frames, often without mentioning the role of acceleration or the need for fictitious forces in non-inertial frames. Others simply stated that the Earth is either inertial or non-inertial, without offering any justification or context to support their claim. Several candidates demonstrated confusion by misusing terms such as "inertia" or by discussing unrelated topics like gravity or planetary motion without connecting them to the frame of reference. In some cases, the explanation lacked structure, or examples were missing or irrelevant. These responses reflected gaps in both conceptual clarity and the ability to apply definitions to real-world contexts, indicating a need for focused instruction and practice on reference frames and their applications.

Image of Weaker Response

Option b:

1. Inertial frame of reference:

Inertial frame of reference is use in, ~~means of~~ means of inertia.

2. non-inertial frame of reference:

non-inertial frame of reference, use in means ~~of~~ of accelerate.

for example: A moving car.

- Earth: generally we view earth as non-inertial frame because of accelerating, moving of earth, but we can also view earth as inertial form of reference.



Suggestions for improvement (Highlight all that apply)

Maximising SLO Achievement	Preferred Pedagogy Used for this SLO	Assessment Strategies
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Any Additional Suggestions:

- Teachers can use comparative charts and concept maps to help students distinguish clearly between inertial and non-inertial frames and to organise their thoughts systematically for writing accurate, well-justified answers.

- Classroom activities involving simulations or demonstrations (e.g., rotating platforms, pendulums) can help visualize the effects of non-inertial motion and clarify when and why fictitious forces arise.

Annexure A: Pedagogies Used for Teaching the SLOs

Pedagogy: Storyboard
Description: A visual pedagogy that uses a series of illustrated panels to present a narrative, encouraging creativity and critical thinking. It helps learners organise ideas, sequence events, and comprehend complex concepts through storytelling.
Example: In a Literature class, students are tasked with creating storyboards to visually retell a novel. They draw key scenes, write captions, and present their stories to the class, enhancing their reading comprehension and fostering their imagination.

Pedagogy: Cause and Effect
Description: This pedagogy explores the relationships between actions and consequences. By analysing cause-and-effect relationships, learners develop a deeper understanding of how events are interconnected and how one action can lead to various outcomes.
Example: In a History class, students study the causes and effects of the Industrial Revolution. They research and discuss how technological advancements in manufacturing led to significant societal changes, such as urbanisation and labour reform movements.

Pedagogy: Fish and Bone
Description: A method that breaks down complex topics into main ideas (the fish) and supporting details (the bones). This visual approach enhances comprehension by highlighting essential concepts and their relevant explanations.
Example: During a Biology class on human anatomy, the teacher uses the fish and bone technique to teach about the human skeletal system. Teacher presents the main components of the human skeleton (fish) and elaborates on each bone's structure and function (bones).

Pedagogy: Concept Mapping
Description: An effective way to visually represent relationships between ideas. Learners create diagrams connecting key concepts, aiding in understanding the overall structure of a subject and fostering retention.
Example: In a Psychology assignment, students use concept mapping to explore the various theories of personality. They interlink different theories, such as Freud's psychoanalysis, Jung's analytical psychology, and Bandura's social-cognitive theory, to see how they relate to each other.

Pedagogy: Audio Visual Resources
Description: Incorporating multimedia elements like videos, images, and audio into lessons. This approach caters to different learning styles, making educational content more engaging and memorable.
Example: In a General Science class, the teacher uses a documentary-style video to teach about the solar system. The video includes stunning visual animations of the planets, interviews with astronomers, and background music, enhancing students' interest and understanding of space.

Pedagogy: Think, Pair, and Share
Description: A collaborative learning technique where students ponder a question or problem individually, then discuss their thoughts in pairs or small groups before sharing with the entire class. It fosters active participation, communication skills, and diverse perspectives.
Example: In a Literature in English class, the teacher poses a thought-provoking question about a novel's moral dilemma. Students first reflect individually, then pair up to exchange their opinions, and finally participate in a lively class discussion to explore different viewpoints.

Pedagogy: Questioning Technique (Socratic Approach)
Description: Based on Socratic dialogue, this method stimulates critical thinking by posing thought-provoking questions. It encourages learners to explore ideas, justify their reasoning, and discover knowledge through a process of inquiry.
Example: In an Ethics class, the instructor uses the Socratic approach to lead a discussion on the meaning of justice. By asking a series of probing questions, the students engage in a deeper exploration of ethical principles and societal values.

Pedagogy: Practical Demonstration
Description: A hands-on approach where learners observe real-life applications of theories or skills. Practical demonstrations enhance comprehension, skill acquisition, and problem-solving abilities by bridging theoretical concepts with real-world scenarios.
Example: In a Food and Nutrition class, the instructor demonstrates the proper technique for filleting a fish. Students observe and then practice the skill themselves, learning the practical application of knife skills and culinary precision.

(**Note:** The examples provided in this annexure serve as illustrations of various pedagogies. It is important to understand that these pedagogies are versatile and can be applied across subjects in numerous ways. Feel free to adapt and explore these techniques creatively to enhance learning outcomes in your specific context.)

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Additionally, we express our gratitude to the esteemed team of reviewers for their constructive feedback on overall performance, better and weaker responses, and validating teaching pedagogies, along with suggestions for improvement.

These contributors include:

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• Dr Naveed Yousuf, CEO, AKU-EB